Farhan Islam

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Education

McMaster University — Hamilton, ON
B.Eng in Electrical Engineering — *Sept 2024 - Apr 2028*

Experience

Body Controls Calibration and System Test Developer (Incoming) — Jan 2027 - Dec 2027
General Motors — *Oshawa, ON*

Low Voltage Electrical Designer — Sep 2026 - Present
McMaster Solar Car — *Hamilton, ON*

- Designing the Motor section schematic of the Electrical System Emulator Board in **Altium Designer**, a **hardware in the loop** platform that emulates the car's **LV** electronics for bench validation of new boards.
- Interfacing an **STM32H563** MCU with dual Mitsuba **M2096C2** controllers over **DB37**, sampling throttle and regen through **3.3 V ADC** and answering **125 kbps CAN** polls with emulated speed and status signals.

Avionics Intern — May 2026 - Aug 2026
Reaction Dynamics — *Longueuil, QC*

- Commissioned a **20-channel DAQ** on a **LabJack T7** with **Mux80** multiplexer expansion, integrating **10** pressure and **10** thermocouple channels under one software-timed acquisition cycle.
- Designed and fabricated a **4-layer** instrumentation **PCB** in **KiCad**, modeling filter and protection response in **LTSpice** and bench-validating the board before DAQ integration.
- Verified signal-chain accuracy by correlating independent shunt-tap measurements with LabJack readings across the full operating range prior to system integration.
- Built a relay-based **electro-hydraulic** control system interfacing a **24 V** wireless radio to a **110 V** pump and solenoid valve, implementing latched relay logic with dual-layer **E-Stop** interlock.

Launch Oversight and Logistics — Sep 2024 - Jan 2025
McMaster Rocketry — *Hamilton, ON*

- Developed **C++** tooling to parse LoRa telemetry logs and automate flight-data visualization, cutting review time to **under 5 minutes per flight**.
- Verified avionics continuity and sensor interfaces with serial-terminal checks, confirming IMU, barometer, and LoRa readiness before launch to **3,187 m** apogee.

Electrical Team Lead — Sep 2023 - Apr 2024
Team 9659, FIRST Robotics Competition — *North York, ON*

- **Led a 5-person electrical team** designing the robot power-distribution architecture, sizing breakers and wire gauges, and building modular **wire harnesses** for full electrical swap-outs in **under 3 minutes** between matches.
- Eliminated mid-match brownouts under **180 A** peak loads by separating noisy motor loads, shortening high-current return paths, and implementing **star grounding** to maintain stable bus voltage.

Projects

Dual-MCU Real-Time Appliance Controller | *C++, ESP32, STM32, Arduino* |

- Engineered a hands-free ultrasonic interface on an **ESP32** using a **3**-sensor HC-SR04 array with staggered mounts, writing custom drivers and inter-MCU **UART** communication for real-time Bluetooth feedback.
- Iterated embedded firmware from single-threshold triggering to cross-validated multi-sensor voting, achieving ±**5 mm** positional accuracy and under **30 ms** feedback latency.

Indoor 3D Spatial Mapping System | *C, Python, MSP432E401Y, VL53L1X, Open3D* |

- Built a portable indoor LiDAR scanner for **under $99 CAD** using an **MSP432E401Y**, **VL53L1X I2C** sensor, and stepper motor, generating 360-degree point clouds across **5** layers at ±**2 mm** calibration accuracy.
- Captured **32** measurements per sweep at **11.25-degree** resolution and wrote a **Python**/NumPy/Open3D pipeline to merge polar data from **5** scan layers into Cartesian point clouds for spatial-mapping analysis.

Technical Skills

Languages: C/C++, Java, Python, MATLAB
Embedded & Firmware: STM32, ESP32, Arduino, MSP432, I2C, SPI, UART, CAN, RTOS/Real-Time Systems
Hardware Design & DAQ: Altium Designer, KiCad, PCB Layout & Routing, LTSpice, LabJack DAQ, Multi-Channel Signal Acquisition, Analog Front-End Design
Power & Controls: Relay Logic, Power Distribution, Wire Harness Design, Safety Interlocks, PID Control
Tools: Git, Soldering, Crimping, Test Procedure Development, AutoCAD